

Dominik Pegler

PERSONAL

Born: October 18, 1982
Place: Vienna, Austria
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ORCID: [0009-0001-2345-1117](https://orcid.org/0009-0001-2345-1117)
Profession: Psychologist
Qualification: Master of Science (MSc)

EDUCATION

2023–2027 **Doctoral Studies, Psychology**

(expected)
University of Vienna

Dissertation Title: *Human-Centered Interpretability in Algorithmic Decision Support and Affective Vision Models*

Advisor: Prof. Frank Scharnowski

Expected completion: July 2027

2021–2023 **Master of Science, Psychology**

University of Vienna

Thesis: *Predicting Psychological Embeddings of Fear-Related Stimuli in a Multidimensional Space Using Deep Neural Networks*

Advisor: Prof. Frank Scharnowski

(graduated with honors)

2015–2021 **Bachelor of Science, Psychology**

University of Vienna

EXPERIENCE

July 2026 – **TU Darmstadt, Germany**

July 2027
Visiting researcher in the *Models of Higher Cognition* group (Prof. Frank Jäkel)

August 2023 – **University of Vienna, Austria**

August 2026
University assistant (Pre-Doc) at the *Methods of Psychology* group (Prof. Frank Scharnowski)

February– June 2023	University of Vienna, Austria Project staff at the <i>Methods of Psychology</i> group (Prof. Frank Scharnowski) working on generalized assignment problems
April–May 2022	University of Vienna, Austria Research internship at the <i>Methods of Psychology</i> group (Prof. Frank Scharnowski) applying machine learning to psychological data
2003–2023	Inject Star, Austria 2015–2023: Customer support and data analytics in manufacturing 2003–2015: Production planning and controlling in manufacturing
August 2001	Alcatel, Austria Summer internship technical support in telecommunications
July 2000	HB Plastik, Austria Summer internship research and development in manufacturing

TEACHING EXPERIENCE

2024-2025	University of Vienna, Austria Theory and Empirical Research 1 (Mind and Brain, Master Psychology) https://ufind.univie.ac.at/en/person.html?id=1001597 Responsibilities: Designed course curriculum and materials, guided students through various programming tutorials on statistics and machine learning, assessed student performance through grading assignments. This course was shortlisted among the top 3 submissions for the teaching award of the Faculty of Psychology at the University of Vienna in June 2026.
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PEER REVIEW

2026	IEEE Transactions on Affective Computing
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MEMBERSHIPS

since 2022	Austrian Psychological Society Österreichische Gesellschaft für Psychologie (ÖGP; www.oegps.at)
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GRANTS AND SCHOLARSHIPS

2026	Marietta Blau Grant Competitive funding by OeAD – Austria’s Agency for Education and Internationalisation – for a 12 month international research stay at TU Darmstadt (Prof. Frank Jäkel) (EUR 21,600).
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COMPUTER SKILLS

Python	PyTorch, pandas, NumPy, Matplotlib, scikit-learn, FastAPI/Flask
Javascript, HTML, CSS	React.js/Next.js, ReactNative
R	Tidyverse, lavaan
SQL	PostgreSQL, SQLite
Rust, C++, julia	Basics
Infrastructure	GNU/Linux, Docker, git
Other	L ^A T _E X, Emacs, JASP, Apache Superset, MS Access, Excel

LANGUAGES

German	Native
English	Fluent
Serbo-Croatian	Basic-Intermediate

INTERESTS

Cognitive Modeling, Reinforcement Learning, Human-Computer Interaction, Cultural Evolution, Interpretability, Computer Vision, Machine Learning, Clinical Research, Problem Solving, New Technologies, Interdisciplinary Projects, Learning, Cycling, Photography, Fiction

PUBLICATIONS (PEER-REVIEWED)

- Zhang, M., Pegler, D., Karner, A., Lor, C., Scharnowski, F., & Melinscak, F. (2026). Mapping mental representations of fear-relevant stimuli through similarity modeling. *Journal of Affective Disorders*. <https://doi.org/10.1016/j.jad.2026.122490>.
- Buchanan, E. M., Cuccolo, K., Heyman, T., van Berkel, N., Coles, N. A., Iyer, A., Peters, K., ... Pegler, D., ... Lewis, S. C. (2025). Measuring the semantic priming effect across many languages. *Nature Human Behaviour*. <https://doi.org/10.1038/s41562-025-02254-x>

PUBLICATIONS (UNDER REVIEW)

- Pegler, D., Steyrl, D., Zhang, M., Karner, A., Arato, J., Scharnowski, F., & Melinscak, F. (2025). SpiderNets: Vision models predict human fear from aversive images. *Under review at Computers in Human Behavior*. <https://doi.org/10.48550/arXiv.2509.04889>
- Pegler, D., Jäkel, F., Steyrl, D., Scharnowski, F., & Melinscak, F. (2026). Unpacking interpretability: Human-centered criteria for optimal combinatorial solutions. *Under review at Computers in Human Behavior Reports*. <https://doi.org/10.48550/arXiv.2603.08856>
- Karner, A., Obenaus, L., Zhang, M., Lor, C., Kostorz, K., Pegler, D., Leopold, M., Melinscak, F., Steyrl, D., & Scharnowski, F. (2024). Visual attributes of spiders associated with aversiveness in spider-fearful individuals: A machine learning analysis. <https://doi.org/10.31234/osf.io/ht2pr>

PUBLICATION (IN PREPARATION)

- Pegler, D., Wu, C., Palminteri, S., & Martins, M. D. (In preparation). Why even accurate experts lose trust: A multi-agent reinforcement learning model. Faculty of Psychology, University of Vienna.
- Pegler, D., Schuster, S., & Verhoef, T. (In preparation). Verbalized sampling preserves diversity at the cost of fidelity in language-model transmission chains. Faculty of Psychology, University of Vienna.

CONFERENCE CONTRIBUTIONS

- Pegler, D., Wu, C. M., Dillion, D., Verhoef, T., Schuster, S. (2026, October). *Inferring transmission preferences in language model agents: From local choices to long-run cultural dynamics*. Simulation of Adaptive Behavior (SAB) Conference, Berlin, Germany. [Poster Presentation]
- Pegler, D., Steyrl, D., Scharnowski, F., & Melinscak, F. (2024, September). *Assessing human interpretability of machine-generated solutions to combinatorial design problems*. Congress of the German Psychological Society (DGPs), Vienna, Austria. [Contributed Talk]